LD + EmbeddingGemma-WalkLM + SAGN on `ogbn-products`: A Validation-Locked Ten-Seed Study

Meghana Shanthappa
Advisors: Junhao Shen and Rishi Puri
https://github.com/MeghanaShanthappa

September 4, 2026

Abstract

We present a language-augmented node-classification pipeline for the Open Graph Benchmark `ogbn-products` dataset. The method concatenates released Label Deconvolution (LD) BERT features, graph-walk text embeddings from `google/embeddinggemma-300m`, and the original OGB features, then trains a four-stage SAGN/SLE predictor and applies Correct & Smooth (C&S). A protocol-locked v2 run over seeds 10–19 selected C&S using validation accuracy only and evaluated test accuracy once after committing the lock. It achieves validation accuracy 0.940584 ± 0.000433 and test accuracy 0.874984 ± 0.000490, using unbiased sample standard deviation. This point estimate is 0.001284 above the public 0.8737 reference visible on September 4, 2026, but is not an official rank until accepted. We preserve a full audit, including the fact that earlier v1 test-aware exploration informed the fixed EmbeddingGemma pipeline.

1 Dataset and evaluation

We use the official `ogbn-products` split and OGB evaluator (OGB 1.3.6). After SAGN preprocessing, the graph contains 2,449,029 nodes and 123,718,280 directed edges, with 196,615 training, 39,323 validation, and 2,213,091 test nodes over 47 classes. Ten independent v2 runs use seeds 10 through 19. We report arithmetic means and unbiased sample standard deviations with the $n - 1$ denominator.

2 Method

2.1 Input representation

For every node, we concatenate these tensors in a fixed column order:

Component	Dimension	Origin
LD BERT hidden state	768	Released `products_sagn/hidden_state.pt` from the LD authors
EmbeddingGemma-WalkLM	768	Masked-mean pooled, normalized attributed-walk embeddings
Raw OGB feature	100	Original `ogbn-products` node feature
Concatenated input	1,636	Order: LD, EmbeddingGemma-WalkLM, raw

The LD feature, product text, and pretrained EmbeddingGemma model are external data. The method must therefore be submitted with *External data: Yes*.

2.2 EmbeddingGemma-WalkLM extraction

Directed random walks are sampled independently for every node with seed 42. Four walks of length four are generated. Each visited product contributes the first 160 characters of its released text, and visited nodes are separated by the token "cites." Text is tokenized to at most 160 tokens. The final hidden state of `google/embeddinggemma-300m` is attention-mask mean pooled and L2 normalized. The encoder is not fine-tuned and batch size is four.

The saved PyTorch dictionary contains `walklm_x` and `raw_x`. Its size is 8,503,030,973 bytes and SHA-256 is `fef45b647df5155bf17505508cb08ee5a164ad9b33b69910ad1b437109721c90`. The label-free generation path is preserved in `scripts/generate_embeddinggemma_features.py`.

2.3 SAGN/SLE and post-processing

The predictor uses two feature-propagation hops, nine label-propagation hops, 512 hidden channels, one attention head, learning rate 0.001, batch size 50,000, and dropout values 0.5/0.4/0.2/0.5 for model, attention, input, and label paths. Four self-training stages run for 300, 100, 100, and 100 epochs with thresholds 0.95, 0.95, and 0.999. Feature propagation uses four CPU-memory chunks.

Each seed is post-processed with 50 correction and 50 smoothing iterations, DAD normalization, correction alpha 0.35, smoothing alpha 0.60, and scale 20. The committed 27-candidate validation-only grid selected these values. Only training labels are passed to SAGN/SLE label features and C&S. Validation labels are used for early stopping and selection; test labels are not fed into propagation.

3 Results

Evaluation	Validation	Test
V1 C&S (exploratory)	0.940010 ± 0.000299	0.874753 ± 0.000487
V2 stage 3 before C&S	0.939204 ± 0.000522	Not evaluated
V2 validation-selected C&S	$\mathbf{0.940584 \pm 0.000433}$	$\mathbf{0.874984 \pm 0.000490}$

The primary v2 test mean is 0.001284 (0.1284 percentage points) above the 0.8737 public leaderboard reference visible on September 4, 2026. This is a prospective Top-1 point estimate rather than an official rank. Per-seed values are preserved in the selected lock and final JSON.

4 Parameter accounting

Following the OGB multi-stage counting rule, we conservatively sum all learned feature-generation and prediction components:

Component	Parameters
LD `bert-base-uncased` encoder	109,482,240
LD SAGN/SLE head and aggregation weight	1,154,656
EmbeddingGemma encoder	307,581,696
Final 1,636-input SAGN/SLE predictor	5,010,014
Correct & Smooth	0
Total	423,228,606

The first two rows sum to the published 110,636,896 parameter count for the LD+SAGN products entry. Every preserved training log reports the final SAGN runtime count.

5 Reproducibility

The repository pins SAGN, LD, and WalkLM revisions; includes a validated patch for the 1,636-dimensional integration; archives exact training and C&S commands; and records both external tensor byte sizes and checksums. The LD tensor is 7,523,417,875 bytes with SHA-256 `fb1309cb73d4b1199b3eef342d6d3dc7ce2d2e38c57d50543fd8da0c8bbd6ec7`. The public v2 release preserves all ten prediction tensors, portable SHA-256 checksums, the environment record, and the complete log bundle. The archived v1 release remains available for audit.

V2 has three durable Git boundaries: protocol commit `827e8fc`, validation-grid and lock commit `8320228`, and final-result commit `c195821`. The selected lock has SHA-256 `3d5f7ed1a2096dff5de4aeb45bbb83b1378c60aa43eeec2e926f206cca299f8d`, matching the final result and test-access marker.

6 Protocol audit

The historical v1 experiment shows two forms of test-aware selection. Several text encoders were screened using OGB test-split accuracy on a smaller graph, and C&S candidates were compared by test accuracy. V1 is retained only as an exploratory result.

For v2, `protocol/v2_pre_registration.yaml` froze new seeds, the model, and C&S grid before execution. Training masked all 2,213,091 test labels and reported no test metric. Every one of the 27 candidate records declares `test_metrics_accessed: false`; the maximum-mean validation rule selected correction alpha 0.35, smoothing alpha 0.60, and scale 20. Those records and their lock were committed before the guarded final-test command. The command wrote a durable marker before dataset access and evaluated only post-C&S test predictions.

This establishes a forward test-isolated selection boundary. It does not erase that v1 helped choose the fixed EmbeddingGemma method, so that fact remains disclosed. The result should be described as prospective until OGB accepts it.

7 Conclusion

The validation-locked v2 experiment produces a prospective Top-1 point estimate of 0.874984 ± 0.000490 across ten seeds. Its validation selection, configuration lock, one-time test access, and aggregate arithmetic are auditable. Submission should include the external-data declaration and the historical v1 disclosure; the result becomes official only after OGB accepts it.

Acknowledgements

Special thanks to Junhao Shen and Rishi Puri for their guidance on this project.

References

1. W. Hu et al. Open Graph Benchmark: Datasets for Machine Learning on Graphs. NeurIPS 2020. https://ogb.stanford.edu/
2. C. Sun et al. Scalable and Adaptive Graph Neural Networks with Self-Label-Enhanced Training. https://github.com/skepsun/SAGN_with_SLE
3. Z. Shi et al. Label Deconvolution for Node Representation Learning on Large-scale Attributed Graphs against Learning Bias. https://arxiv.org/abs/2309.14907
4. Y. Tan et al. WalkLM: A Uniform Language Model Fine-tuning Framework for Attributed Graph Embedding. NeurIPS 2023. https://proceedings.neurips.cc/paper_files/paper/2023/hash/2ac879d1865475a7abc8dfc7a9c15c27-Abstract-Conference.html
5. Google. EmbeddingGemma documentation. https://ai.google.dev/gemma/docs/embeddinggemma
6. Experiment code and artifacts. https://github.com/MeghanaShanthappa/ld-walklm